Toward Highly Flexible Inter-User Calibration of Myoelectric Control Models With User-Defined Hand Gestures

Yangyang Yuan, Zihao Chen, Jionghui Liu, *Graduate Student Member, IEEE*, ChihHong Chou, Chenyun Dai, *Member, IEEE*, and Xinyu Jiang

***Abstract*—Myoelectric control models enabling accurate hand gesture recognition via electromyography (EMG) have attracted increasing attentions in rehabilitation robotics. Adapting pre-trained models to new users is a main challenge in real world applications due to the inter-user different EMG characteristics. Most previous transfer learning approaches employed a rigid model calibration process, usually in a supervised manner with ground truth labels, or in an unsupervised manner but still requiring users to perform pre-defined hand gestures to update model parameters. We argue that such a rigid model calibration process lacks flexibility and limit the translation of myoelectric control into real world practice. In this work, we gradually "flexibilize" the standard model calibration process toward a highly flexible version, which does not require the labels of calibration data, and can be performed on only a subset of pre-defined hand gestures or even unknown user-defined hand gestures. We identify those key components contributing to the performance difference along the way. Compared with the supervised method, the unsupervised model calibration even contributed to a 10% improvement ($p < 0.05$) in case where only a subset of gesture categories were available for model calibration. Moreover, the unsupervised model calibration achieved a highest recognition accuracy of 86.57% using unknown user-defined gestures, with no significant difference compared to the accuracy with pre-defined gestures ($p > 0.05$).**

***Index Terms*—Myoelectric control, EMG, transfer learning, domain adaptation.**

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Yangyang Yuan is with the School of Information Science and Technology, Fudan University, Shanghai 200433, China (e-mail: yyyuan22@m.fudan.edu.cn).

Zihao Chen is with the School of Electrical and Computer Engineering, Georgia Institute of Technology, Atlanta, GA 30332 USA (e-mail: zchen959@gatech.edu).

Jionghui Liu is with the Institute of Science and Technology for Brain-Inspired Intelligence, Fudan University, Shanghai 200433, China (e-mail: jhliu22@m.fudan.edu.cn).

ChihHong Chou and Chenyun Dai are with the School of Biomedical Engineering, Shanghai Jiao Tong University, Shanghai 200241, China (e-mail: Knives.chou@gmail.com; chenyundai@sjtu.edu.cn).

Xinyu Jiang is with the School of Informatics, The University of Edinburgh, EH8 9BT Edinburgh, U.K. (e-mail: xinyu.jiang@ed.ac.uk).

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I. Introduction

HUMAN-MACHINE interaction (HMI) system detects neural commands through collecting biological signals from the human body, and decodes them into machine languages enabling users to control electronic devices [1], [2]. Surface electromyography (sEMG) containing rich information on users' motion intention is widely used due to its high signal-to-noise ratio and non-invasive signal acquisition [3]. Nowadays, HMI system has become a research hotpot for their wide applications in prosthesis control [4], [5], [6] and motor rehabilitation [7], [8], [9].

In practical use, the inter-user differences of sEMG characteristics limit the application of myoelectric control models on a new user [10]. Transfer learning, which can transfer the knowledge learned from the data of other users (the source domain) to the data of a new user (the target domain), provides a possible solution to this problem [11], [12], [13]. Côté-Allard et al. [14] reduced the training burden and improved the model generalization by transferring the weights of the existing model with progressive neural networks (PNN). In the same scenario, Kim et al. [15] trained a convolutional neural network (CNN) and fine-tuned the model for the new user, improving the classification accuracy by 23.2%. In another study of Chen et al. [16], both CNN-only network and CNN + long short-term memory (LSTM) network with the transfer learning were implemented. Previous studies have proved the effectiveness of transfer learning to calibrate a pre-trained myoelectric control model on a new user.

However, most previous studies selected a supervised transfer learning algorithm to optimize their models, where the models are fine-tuned with labeled data to learn the data distribution of a new user. To free model calibration from labeled data for more convenient application, many studies have adopt unsupervised model calibration methods. In Khushaba's research [17], canonical correlation analysis (CCA) was utilized to implement style-independent feature transformation. The recognition model overcame individual differences with accuracies over 83% across multiple users. Based on the algorithm, Xue et al. [18] combined CCA model with optimal transport (OT) for cross-user model optimization. The proposed CCA-OT method improved the recognition accuracy by 3.81% compared with CCA-only method, and 43.54% compared with benchmark model. By contrast, Zhang et al. [19] applied

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domain adaptation methods to improve the model performance in cross-user scenarios. They developed a model based on the network of maximum classifier discrepancy (MCD), achieving an accuracy of 94.59% on the sports activities data set (DSADS). In the follow-up study, Zhang [20] argued that the previous studies of domain adaptation methods in sEMG control are only based on single-source domain adaptation, ignoring the sEMG distribution difference between training users. Based on this, Zhang proposed a multi-source synchronize domain adaptation framework. The ablation experiment showed that, with multi-source domain adaptation, the recognition accuracy was significantly improved from 75.19% to 87.38%. In a recent research conducted by Wang et al. [21], several classic domain adaptation methods were compared. Through the validation on three open-source datasets, transfer component analysis (TCA) and manifold embedded distribution alignment (MEDA) algorithms achieved relatively better performance, particularly in isolated scenes. In general, previous es have demonstrated that unsupervised learning methods are highly effective in cross-user model calibration and made significant contributes to the comparison and improvement of unsupervised learning algorithms. Nevertheless, these researches still assume an ideal but not realistic application situations where all class categories were available during model calibration. This will cause inconvenience in practical use because the samples collected during daily use are usually unlabeled and sometimes even undefined during model training. An ideal and highly flexible model calibration algorithm should be run without the gesture labels, and can be performed on only a subset of pre-defined hand gestures or even undefined new hand gestures.

In response to the aforementioned issues, in this work, we aim to examine the feasibility to develop a flexible model calibration framework which does not require the labels of calibration data, and can be performed on only a subset of pre-defined hand gestures or even unknown user-defined hand gestures. We investigated the model calibration problem in four different levels. First, we implemented the most basic types of models, namely user-specific models, pre-trained user-independent models, and pre-trained then (supervised) fine-tuned models, to serve as baseline performances. Second, multiple unsupervised model calibration algorithms were implemented to make the model calibration process free of ground truth labels. Third, we evaluated the performance of both supervised and unsupervised model calibration methods using only a subset of hand gestures involved in model training. Fourth, we further perform unsupervised model calibration using user-defined hand gestures unseen during model training. Different from previous studies [21], [22] aiming to make the model calibration process from certain requirements (e.g., data labels or pre-defined gesture categories), in this work, we gradually "flexibilize" the standard model calibration process toward a highly flexible version. We identify those key components contributing to the performance difference along the way. All the calibration methods were validated on the data from 20 subjects in an open access dataset Hyser [23]. We found that unsupervised model calibration achieved comparable model performance to the supervised method when all gesture categories were available, and even contributed to a higher robustness using only a subset of gesture categories or unknown hand gestures. We expect our work to facilitate the real world applications of myoelectric control models by testing the flexibility limits of the model calibration process.

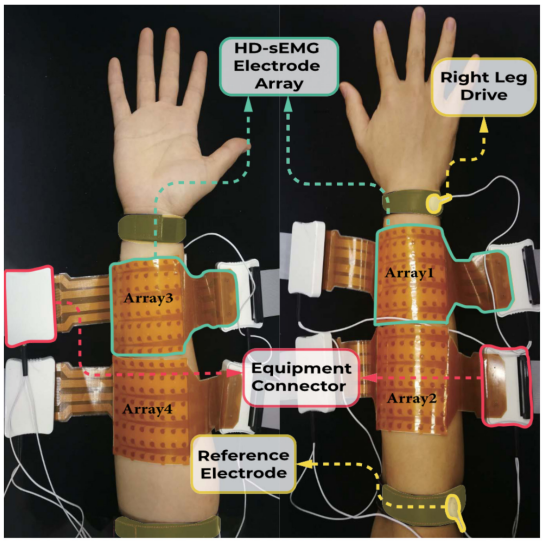


Fig. 1. Electrode setup in experiment.

II. Materials

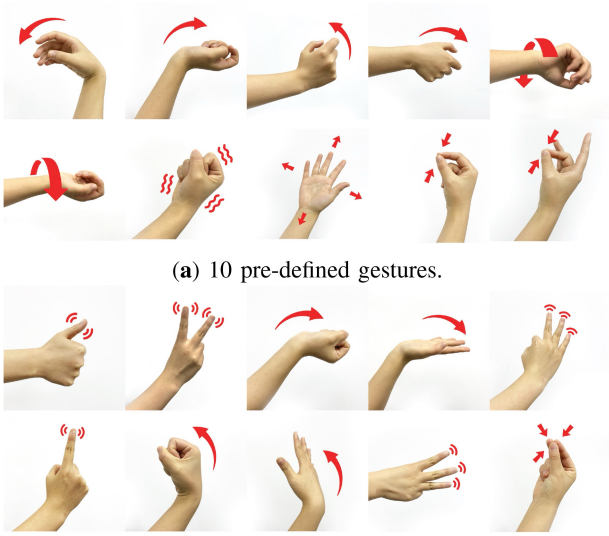
A. Subjects

The dataset (Hyser dataset) used in this study has been open sourced in our previous work [23]. Specifically, we used the sEMG data from the Pattern Recognition subset of the dataset Hyser, including 20 subjects with 12 males and 8 females aged between 22 and 31 years. All subjects are right-handed. Before the experiment, all the subjects were well informed of the details of the experiment. The study was supervised and approved by the ethics committee of Fudan University (approval number: BE2035). Here we briefly introduce the data acquisition process.

B. Data Acquisition

During the experiment, we used four 8×8 electrode arrays mounted on both extensor and flexor muscles (two arrays for each) of the forearm to collect 256-channel sEMG signals, as shown in Fig. 1. In addition, the right leg drive and reference electrodes were placed on the head of the ulna and the elbow respectively. The signals were acquired by the Quattrocento system (OT Bioelettronica, Torino, Italy), sampled at a rate of 2048 Hz. The amplifier gain was 150 and the ADC resolution was 16 bits.

The data from the Pattern Recognition subset was collected in a quiet room. The subjects were required to sit in front of a screen displaying the experimental instructions and perform 34 hand gestures under the guidance. Each gesture included 2 trials. Each trial consisted of three 1s dynamic tasks (reaching the target hand gesture from resting state) and one 4s holding task (reaching the target gesture and holding for 4 seconds). To avoid the muscle fatigue, a 2s inter-task resting period and a 5s



(**a**) 10 pre-defined gestures.

(**b**) 10 unknown gestures.

Fig. 2. Gestures involved in this study.

inter-trial resting period were provided during the experiment, as shown in Fig. 2. If the subject performed a wrong gesture or missed a certain task, they would inform the experiment assistant. Then, these tasks would be removed from the dataset according to the experiment records. In the final dataset, we excluded 2.30 ± 2.71 dynamic tasks and 0.85 ± 1.05 gesture holding tasks on average. Only dynamic tasks were used in this work since they are more commonly performed in real world scenarios.

C. Data Selection

In this study, we selected 10 of the 34 gestures as the pre-defined hand gestures involved during model training and another 10 gestures as unknown gestures (shown in Fig. 2). Specifically, the 10 pre-defined gestures were used for recognition tasks. In addition, to increase the convenience for user calibration, this study allows users to use arbitrary gestures for calibration that does not belong to the targeted 10 pre-defined gestures. Accordingly, the 10 unknown gestures were selected to the freedom of model calibration procedures. All the 20 gestures were commonly used in our daily life, involving various motions states of fingers and wrist.

III. Methods

A. Data Preprocessing

The raw sEMG signal was filtered with a 10 Hz high pass filter and a 500 Hz low pass filter sequentially (both Butterworth filter, bidirectionally with 8-order for each direction). Then, a group of notch filters were employed to attenuate the interference of the power frequency at 50 Hz and all its harmonic components below 400 Hz.

B. Backbone Model

The neural network serving as a backbone was composed of a VGGNet-16-BN [24] feature extractor module and a linear decision making module. The framework of the whole network is shown in Fig. 3(a). The sEMG example size of each dynamic task for model training is 256 (channels) × 2048 (2048 Hz × 1s). To enable the network to extract temporal information using convolution kernels, we reshaped the 2048-dimension into 64 × 32. Accordingly, the filtered sEMG waveforms with a shape of 256 × 64 × 32 were directly used as the input of the end-to-end network. The classification loss function, denoted as $\mathcal{L}_{class}$, was defined as

$$\mathcal{L}_{class} = -\left[y_n \cdot log(f(x_n)) + (1 - y_n) \cdot log(1 - f(x_n))\right], \quad (1)$$

where $f(x_n)$ refers to the network output for sample x_n, and y_n refers to its one-hot real label. This network was used as the backbone model in our following analyses.

C. Baseline 1: User-Specific Model Trained From Scratch

To develop a user-specific model, data of the 10 pre-defined hand gestures from the same user in the same data collection session were used in both training and testing the model. For each hand gesture, the first 3 examples were used to train a model from scratch, with the remaining examples used to test the model performance. The performance were evaluated twice using data collected from two sessions separately (the same for all following analyses). The average accuracy of the two sessions was reported. The framework of the user-specific model is presented in Fig. 3(a).

D. Baseline 2: Pre-Trained User-Independent Model

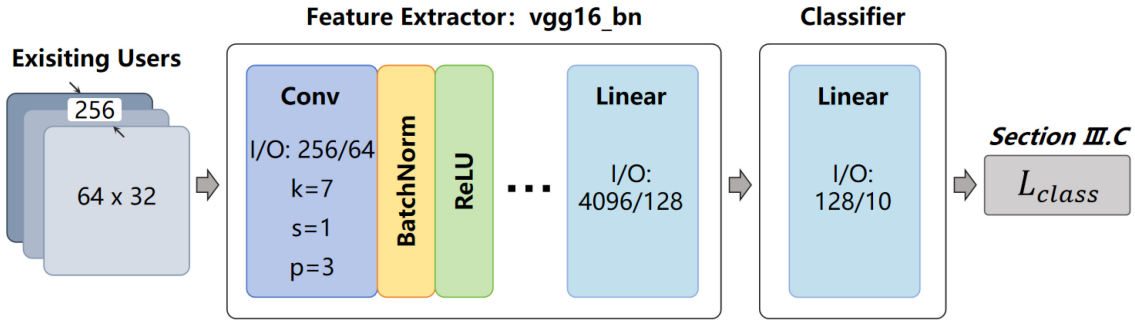
To develop a user-independent model, the leave-one-user-out cross-validation strategy was applied. Data from each user were used as the testing data in turn, with data from the other 19 users used to pre-train a model. The pre-trained model was applied directly to the testing user without any model calibration process. Only the 10 pre-defined hand gestures were used in this setting.

E. Supervised Model Calibration

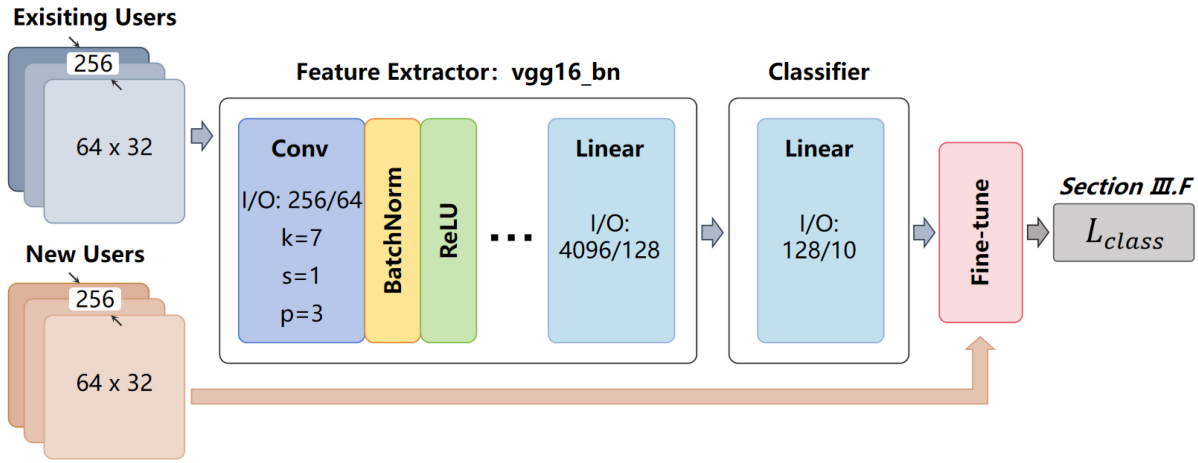
Given a pre-trained user-independent model presented in the above setting, we aim to calibrate the pre-trained model in a supervised way. We allocated 3 examples for each hand gesture from the testing user into the calibration dataset to fine-tune the model, by further performing gradient descent on model parameters (50 epochs). Only the 10 pre-defined hand gestures were used in this setting. The overall framework of supervised model calibration is presented in Fig. 3(b).

F. Unsupervised Model Calibration

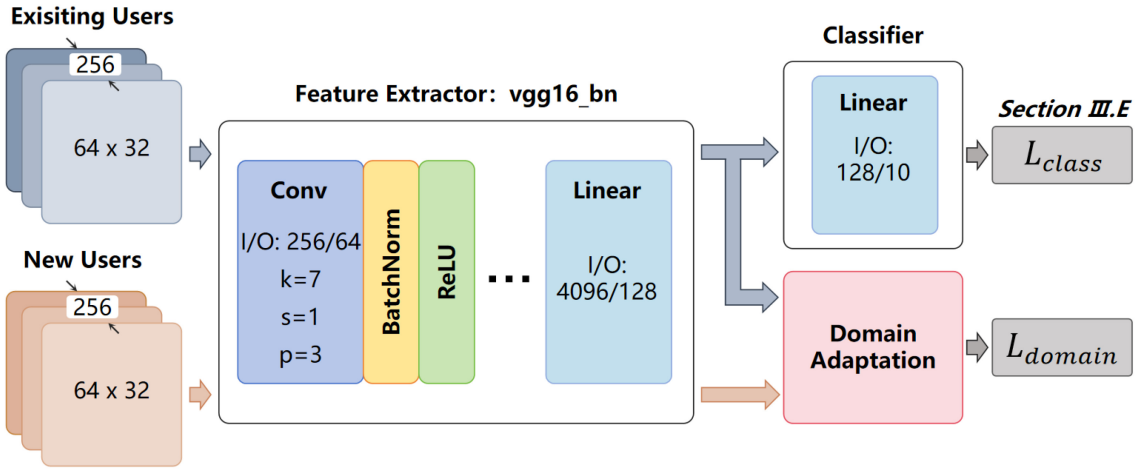
In this setting, we aim to calibrate a pre-trained model in an unsupervised way. The same as the above setting, data from other 19 users were allocated to the training dataset (source domain), 3 examples for each hand gesture in a session form the testing user were allocated to the calibration dataset (target domain), with the remaining examples as the final testing dataset. The only difference is that the ground truth labels in the calibration dataset are not available. Only the 10 pre-defined hand gestures were used in this setting.



(**a**) User-Specific Model. Highest gesture recognition accuracy achieves 77.23%.



(**b**) Supervised Model Calibration. Highest gesture recognition accuracy achieves 87.88%.



(**c**) Unsupervised Model Calibration. Highest gesture recognition accuracy achieves 87.16% with pre-defined gestures and 86.57% with unknown gestures.

Fig. 3. The framework of the models used in this study. k, s and p respectively denote the kernel, stride and padding size of the Convolution Layer. I/O denotes the input/output channel number of the Convolution Layer. The italics above the L_{class} label indicate the section containing the detailed introduction of the corresponding model.

Unsupervised model calibration aims to adapt the distribution of output data in a latent space [25], as illustrated in Fig. 3(c). Therefore, a measure of the mismatch between the distributions in the source domain (denoted as $\mathcal{D}_s$) and the target domain (denoted as $\mathcal{D}_t$) is required as another term of loss function, denoted as $\mathcal{L}_{domain}$. A total of five domain adaptation techniques, namely (1) conditional adversarial domain adaptation (CDAN) [26], (2) domain-adversarial neural networks (DANN) [27], (3) maximum mean discrepancy (MMD) [28], (4) Wasserstein MMD (WMMD) [29], and correlation alignment (CORAL) [30] were employed to measure and adapt the distribution mismatch, and their performances were systematically compared. Considering CORAL contributed to the best performance (see Results Section for details), here we particularly introduce the principle of CORAL. The details of other techniques can be found in [26], [27], [28], [29].

CORAL aims to minimize the differences between the second-order statistics of source and target domains, defined by the following equation:

$$\mathcal{L}_{domain} = \frac{1}{4d^2}\|C_s - C_t\|_{\mathcal{F}}^2, \tag{2}$$

where d denotes the dimensionality of the feature vector in the latent space, $\|\cdot\|_{\mathcal{F}}$ represents the matrix Frobenius norm. C_s and C_t denote the covariance of the source and target domains, respectively, given by the following equations:

$$\mathcal{C}_s = \frac{1}{N_s - 1}\left(D_s^T D_s - \frac{1}{N_s}\left(\mathbf{1}^T D_s\right)^T\left(\mathbf{1}^T D_s\right)\right)^2 \tag{3}$$

$$\mathcal{C}_t = \frac{1}{N_t - 1}\left(D_t^T D_t - \frac{1}{N_t}\left(\mathbf{1}^T D_t\right)^T\left(\mathbf{1}^T D_t\right)\right)^2, \tag{4}$$

where N_s and N_t denote the number of samples in the source and target domains, respectively; D_s and D_t denote the data matrix in the source and target domains, respectively; $\mathbf{1}$ denotes a vector with all its elements equal to 1. With $\mathcal{L}_{domain}$, the network would learn to adapt the distribution match between both domains at the same time when learning to classify different hand gestures. The overall framework of supervised model calibration is presented in Fig. 3(c).

G. Model Calibration With a Subset of Pre-Defined Hand Gestures

In this setting, we aim to calibrate a pre-trained model using only a subset of hand gestures. Specifically, we randomly selected 2 and 6 out of the 10 pre-defined hand gestures (for each selected gestures, only the first 3 examples were used to calibrate the model). The random selection process was repeated 3 times for each testing user. Both supervised and unsupervised model calibration methods were performed in this setting. All other configurations were the same as other calibration settings.

H. Model Calibration With Unknown User-Defined Hand Gestures

In this setting, we aim to calibrate a pre-trained model using user-defined hand gestures which are unknown during model training. Specifically, we randomly selected 2 and 6 out of the 10 unknown hand gestures (for each selected gestures, only the first 3 examples were used to calibrate the model). The random selection process was repeated 3 times for each testing user. All other configurations were the same as other calibration settings.

I. Statistical Analysis

The gesture recognition performance of different model calibration methods were compared using statistical analysis. Since the results of model accuracy did not follow a Gaussian distribution, the nonparametric Friedman test was applied to multiple comparisons, and Wilcoxon Signed-Rank test to paired comparison. The Nemenyi post-hoc test was used for further comparisons if necessary. The Bonferroni-Holm correction was also applied to avoid multiple comparison errors. Only p-values were reported, and a significance was observed if $p < 0.05$.

TABLE I
GESTURE RECOGNITION ACCURACY FOR BASELINE MODELS

Model	user-specific	user-independent
Accuracy	80.25±11.95	77.23±12.69

TABLE II
GESTURE RECOGNITION ACCURACY OF THE SUPERVISED MODEL CALIBRATION METHOD USING DIFFERENT NUMBER OF PRE-DEFINED GESTURE CATEGORIES

Number of Gesture	10	6	2
Accuracy	87.88±8.17	85.63±10.15	72.13±13.40

IV. RESULTS

A. Results of Baseline Models

The hand gesture classification of baseline models are presented in Table I. User-specific models trained from scratch using only the data from the target user contributed to an average accuracy of 80.25% ± 11.95%. User-independent models pre-trained using data from other users without any model calibration contributed to an average accuracy of 77.23% ± 12.69%. The paired Signed-Rank test shows that the two baseline methods have no significant difference ($p = 0.534$). This result may indicate that although the accuracy of the user-independent model may slightly decrease compared to the user-specific model, the performance can be still close if a large amount of subject samples can be used for inter-user model training.

B. Results of Supervised Model Calibration Methods

Table II presents the classification accuracy of the supervised model calibration method, using different number of pre-defined gesture categories. When using all 10 pre-defined hand gestures during the model calibration process, the accuracy of pre-trained model increased from 77.23% ± 12.69% to 87.88% ± 8.17% by 10.65% ($p < 0.05$). When using 6 out of 10 pre-defined hand gestures during the model calibration process, the accuracy of pre-trained model increased from 77.23% ± 12.69% to 85.63% ± 10.15% by 8.40% ($p < 0.05$), but no significance was found compared to using all 10 pre-defined gestures ($p > 0.05$). However, when only 2 out of 10 pre-defined hand gestures, the accuracy dropped from 77.23% ± 12.69% to 72.13% ± 13.40%. The overall results demonstrate that the performance of supervised model calibration method can be improved using a certain amount of pre-defined gestures, but is not flexible and robust enough with a small subset of classification categories.

C. Results of Different Unsupervised Model Calibration Methods

Table III presents the accuracies of different unsupervised model calibration methods. Compared with the accuracy of 77.23% ± 12.69% using the baseline user-independent model without domain adaptation, CDAN, DANN, MMD, WMMD

TABLE III
GESTURE RECOGNITION ACCURACY OF DIFFERENT UNSUPERVISED MODEL CALIBRATION METHODS

Transfer Method	user-independent (baseline)	CDAN	DANN	MMD	WMMD	CORAL
Accuracy	77.23±12.69	67.50±14.93	78.53±14.85	78.62±12.20	87.40±10.65	87.16±11.21

TABLE IV
GESTURE RECOGNITION ACCURACY OF THE CORAL-BASED UNSUPERVISED MODEL CALIBRATION METHOD USING DIFFERENT NUMBER OF PRE-DEFINED GESTURE CATEGORIES

Number of Gesture	10	6	2
Accuracy	87.16±11.21	85.10±11.90	82.86±12.24

TABLE V
GESTURE RECOGNITION ACCURACY THE CORAL-BASED UNSUPERVISED MODEL CALIBRATION METHOD USING DIFFERENT NUMBER OF UNKNOWN GESTURE CATEGORIES

Number of Gesture	10	6	2
Accuracy	86.57±12.69	84.75±8.21	81.75±8.75

and CORAL contributed accuracies of 67.50% ± 14.93%, 78.53% ± 14.85%, 78.62% ± 12.20%, 87.40% ± 10.65%, and 87.16% ± 11.21%. The Friedman test shows a significant difference across all these methods ($p < 0.05$). Specifically, WMMD and CORAL achieved comparable performances with each other and a significant improvement on the baseline model without using domain adaptation ($p < 0.05$). Considering WMMD require longer computation time during the calibration process, we employed CORAL in all our following analyses.

D. Results of CORAL-Based Unsupervised Model Calibration Methods

Table IV presents the classification accuracy of CORAL-based unsupervised model calibration method, using different number of pre-defined gesture categories. If all 10 pre-defined hand gestures were available for model calibration, the accuracy of pre-trained model significantly increased from 77.23% ± 12.69% to 87.16% ± 11.21% by 9.93% ($p < 0.05$). Such accuracy is comparable with the supervised calibration method (87.88% ± 8.17%) in the same setting ($p = 0.807$). When using 6 out of 10 pre-defined hand gestures during the model calibration process, the accuracy of pre-trained model significantly increased from 77.23% ± 12.69% to 85.10% ± 11.90% by 7.87% ($p < 0.05$). When only 2 out of 10 pre-defined hand gestures, the accuracy still increased from 77.23% ± 12.69% to 82.86% ± 12.24% by 5.63% although no significance was found ($p = 0.144$). Compared with the supervised model calibration method using only 2 pre-defined hand gestures, the unsupervised method achieved a higher accuracy (82.86% ± 12.24% vs. 72.13% ± 13.40%) ($p < 0.05$).

Moreover, even when all pre-defined hand gestures were not available and users only performed unknown hand gestures during the model calibration process, the CORAL-based unsupervised method could still improve the accuracy of the pre-trained model. As presented in Table V, with 10, 6 and 2 unknown hand gestures, CORAL-based method contributed to accuracies of 86.57% ± 12.69%, 84.75% ± 8.21%, and 81.75% ± 8.75%, respectively. The accuracy of Coral-based calibration using unknown gestures is slightly (approximately 1%) lower than that of using the same number of known gestures, but all much higher than the baseline accuracy of the pre-trained model without domain adaptation (77.23% ± 12.69%), demonstrating the excellent robustness of CORAL-based unsupervised method.

V. DISCUSSION

In this work, we systematically explored the feasibility and potentials of a highly flexible calibration framework for myoelectric control models. We flexibilize the common model calibration framework by removing the 2 demands: (1) the demand on ground truth labels of the calibration data, and (2) the demand on calibration data of all or most of pre-defined classification categories. As for the first demand, previous studies have explored unsupervised calibration methods for myoelectric control models. The advantages of unsupervised methods is that users do not need to follow a rigorous data collection session synchronized with the instructions to associate each data sample to a class label. The unsupervised method could gradually update the model using saved testing data without labels. However, previous studies on unsupervised model calibration still assume an ideal application scenario where (1) users perform all pre-defined hand gestures, (2) the number of collected data samples are balanced across all classes, and (3) users does not perform other random hand gestures. With the above three assumptions, the requirements on the data collection session for unsupervised methods is still quite rigorous even if the ground truth labels are not necessary. In this work, we remove all these demands in the model calibration process. The robustness of our model calibration framework is verified in various settings.

In company with the calibration demands, calibration complexity (computational resources for the model training of machine learning algorithms) and recognition accuracy are two trade-off factors required to be fully considered according to the specific practical scenarios. The performance of different models evaluated by the three factors is illustrated in Fig. 4. First, the baseline user-independent model without additional calibration has the lowest calibration complexity and demands. However, the accuracy reaches as low as 77.23%, which may not satisfy the practical requirement of scenarios requiring high recognition accuracy. Second, the unsupervised calibration model can realize a very high accuracy with a moderate calibration demands and computational complexity. This model even allows for the use of arbitrary unknown gesture calibration without the need for using pre-defined gestures, greatly improving the convenience of practical use. Third, although the supervised pre-defined calibration model can

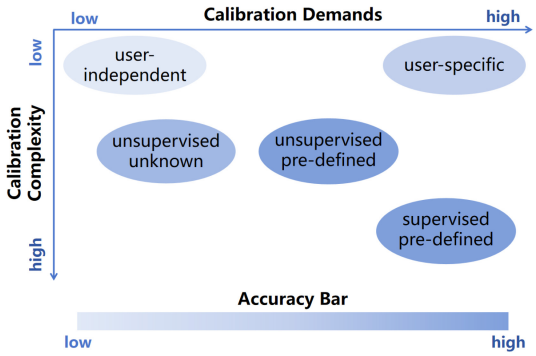


Fig. 4. Model performance evaluated by three factors. The horizontal axis represents the rigorousness of the calibration demands, involving data label and gesture categories. The vertical axis represents the simplicity of the calibration procedures, measuring by training duration. The color of the ellipse reflects the highest gesture recognition accuracy achieved by corresponding model.

achieve high recognition accuracy, its calibration demands are extremely high, which requires a large amount of pre-defined data with the labels. Moreover, the required computational resources of model training is the highest.

Conventional supervised model calibration methods mainly fine-tune the parameters of the pre-trained neural network by further applying gradient descent using the calibration data. In supervised model calibration, models mainly learn to discriminate patterns of different classes by modelling the data distribution of each single class. Such data distribution corresponding to a certain class cannot be estimated and modelled without relevant calibration data. The above demand is a main limitation of supervised calibration method, in spite of the high accuracy, given data from all class categories. Unsupervised methods, by contrast, do not model the distribution of data in each single class, but aim to adapt the overall data distribution between the target and the source domains. Each sample, regardless of its ground truth label, could contribute to a better description on the overall data distribution. Additionally, certain important characteristics of EMG from a specific user in the target domain, e.g., features related to EMG amplitude, show a consistent shift from the source domain for all gesture categories. For example, a target user with lower force level would generate consistently lower EMG amplitude compared with the source domain when performing all hand gestures. Such consistent bias on data distribution can be captured on only a subset of hand gestures, or even other unknown random hand gestures.

From the aspect of classification accuracy, the supervised learning/calibration methods are generally considered a better solution compared with the unsupervised ones. In other words, the unsupervised model calibration methods usually contribute to a much lower classification accuracy, and are considered as a compromise of supervised methods when the ground truth labels are not available. Based on our analysis, we conclude that when the data distribution across populations is similar, unsupervised methods might contribute to even higher classification accuracy compared with supervised methods and are more robust with a subset of class categories. However, previous studies have shown that age or ethnicity may have a certain impact on the overall accuracy of gesture recognition [31]. When the source domain and target domain come from populations of different ethnicities or age groups, the accuracy of the cross-subject model may decline. This is due to potential differences in the overall distribution of sEMG data caused by variations in age or ethnicity. However, due to limitations in data collection, this study did not cover a wide range of age groups or ethnicities. Nevertheless, these demographic differences may affect the performance of different models. For instance, unsupervised learning algorithms are calibrated based on the overall data distribution between training and testing data. If the target testing population significantly differs demographically from the training population, the accuracy may decrease after model transfer. In contrast, supervised learning calibration is mainly based on data from the target testing population. Therefore, the degree of decrease in model performance may be relatively small.

In addition to diverse model calibration methods explored in this work, the pre-training step using a large amount of data from other users is also quite important to improve the generalization performances. Without any calibration data from the target user to personalise a pre-trained model, the user-independent model achieved a comparable accuracy (77.23%) with the user-specific model trained merely on data from the target user (80.25%). The pre-training step enables the myoelectric control models to learn general knowledge on EMG characteristics and data distribution, largely reducing the frustratingly heavy data collection burdens on each new user. The efforts to collect pre-training dataset are once for all, and can be reused by all new users with only a few random gesture samples required. Given that the proposed unsupervised calibration method only needs additional calculation of the second moment of the samples, it imposes little additional burden on time consumption or computational efficiency compared to the original training process. With an extremely large pre-training dataset, the number of required calibration data samples can be further reduced and the performance of a calibrated model can be even improved.

In previous studies [32], [33], [34], [35], [36], [37], various supervised and unsupervised algorithms have been proposed to improve the classification accuracy of models. Some of these methods achieved significant improvements in accuracy for a more efficient model, whereas others devoted to unsupervised methods for more convenient calibration. In previous research, Rahimian et al. [32] proposed a novel Few-Shot learning-Hand Gesture Recognition (FS-HGR) architecture, improving the accuracy to 81.29% on new subjects in 50 gesture recognition tasks. However, the classification accuracy dramatically dropped to 73.36% on new gestures. In the similar scenario, our model demonstrated higher robustness with accuracy declining by less than 1% (87.16% to 86.57%), which could be attributed less gesture categories involved in recognition (with only 10 gestures in our study). For unsupervised methods, Campbell et al. [37] extended adaptive domain adversarial neural network (ADANN) on new users, achieving an average accuracy of 91.5%. This model outperformed ours in classification accuracy, likely due to the use of more samples during

training and calibration than ours (9 samples v.s. 3 samples). Additionally, the aforementioned studies used data from all gesture categories for calibration, our results suggest that the proposed unsupervised calibration strategy can offer greater flexibility (e.g., using unknown user-defined gestures) and require lower cost (e.g., using a subset of gesture categories).

Although the model proposed in this study achieved good results in cross-user scenarios, one limitation is that all results were analyzed offline. In one recent published study [34], the same database and the same transfer learning framework were used to achieve real-time cross-user gesture recognition. Here, we briefly introduced the real-time framework. The efforts to collect pre-training dataset are once for all, and can be reused by all new users with only a few random gesture samples required. Given that the proposed unsupervised calibration method only needs additional calculation of the second moment of the samples, it imposes little additional burden on time consumption or computational efficiency compared to the original training process. Since we mainly focused on the flexibility in data requirement for different calibration strategies, our hypothesis has only been validated in offline scenarios. In our future research, we will focus on further optimizing the model for online scenarios, and devote to embedding it into hardware systems. Another limitation is that all the validations were conducted using the commonly used commercial EMG product, the Quattrocento system (OT Bioelettronica, Torino, Italy). The electrode configurations used high-density arrays covering the forearm. Different acquisition systems, electrode configurations, and collection environments may affect the accuracy of recognition results.

VI. Conclusion

In this work, we gradually flexibilize the conventional model calibration framework to a highly flexible version. We evaluated the effects of different components on the performance of calibrated models along the way. Our flexible model calibration framework does not require the ground truth labels of calibration data, and can be performed on only a subset of pre-defined hand gestures or even unknown user-defined hand gestures. Another important finding of our work is that, the unsupervised model calibration even contributed to a better performance and higher robustness compared with the supervised method, using only a subset of gesture categories or unknown hand gestures. We expect our work can promote the translation of myoelectric control into more real world applications.

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